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End Semester Examination December- 2025

Name of the Course: **B.Tech.**Name of the Paper: **Heat Transfer**Semester: **V**Paper Code: **TME501****Time: 3 Hours****Maximum Marks: 100****Note:**

- (i) All questions are compulsory
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) Total marks in each question are twenty
- (iv) Each question carries 10 marks

Q1 (20 marks)		
(a)	Derive the expression for the rectangular fin in the case of heat dissipation from FIN insulated at tip.	CO3
(b)	The surface of a vertical cylinder, 9 cm outer diameter and 1.5 m long is in contact with saturated steam at atmospheric pressure. The surface of cylinder is maintained at 40° C by flowing water through it. Calculate the rate of heat transfer to water and rate of condensation of steam. Assume suitable properties of water.	CO6
(c)	A thin metal plate is kept insulated on one side and exposed to the sun on other side. The plate absorbs solar energy at a rate of 500W/m ² and dissipates it by convection into the ambient air at 300 K with a convection coefficient 20W/m ² K and by radiation into a surrounding area which can be considered as a black body at 280 K. The emissivity of the surface is 0.9. Calculate the equilibrium temperature of the plate.	CO6
Q2 (20 marks)		
(a)	Explain free and forced convection in detail and describe the boundary layer development phenomenon over a flat plate	CO2
(b)	A straight fin of rectangular profile is constructed of stainless steel (k= 16 W/mK) and has a width of 5 cm and thickness of 2.5 cm. The base temperature is maintained at 100° C and fin is exposed to a convection environment at 20° C with h= 47W/m ² K. Calculate the heat loss by the fin of one meter length.	CO3
(c)	Explain how the surface area is calculated in heat exchangers and discuss the concept of fouling.	CO4
Q3 (20 marks)		
(a)	Derive the relation for heat conduction through a hollow sphere and explain the concept of logarithmic mean area.	CO2
(b)	Explain the following dimensionless numbers with their physical significance (i) Prandtl number (ii) Reynold's number (iii) Grashof number (iv) Nusselt number	
(c)	Two concentric spheres of diameter 20 cm and 30 cm separated by air at one atmospheric pressure. The surface temperatures of the two spheres are 320 K and 280 K respectively. Determine the heat transfer by natural convection from the inner sphere to the outer sphere. Use the relation $Nu = 0.228Ra^{0.226}$ Properties at mean film temperature are $K = 0.026W / mK$ $Kinematic\ viscosity = 1.57 \times 10^{-5} m^2 / s$ $Pr = 0.712$	CO2 CO3
Q4 (20 marks)		

(a)	Derive the expression for LMTD for parallel flow heat exchanger.	CO2
(b)	Calculate the overall heat transfer coefficient U_o based on outer surface of a pipe with an inner diameter 3 cm and outer diameter 4 cm. Thermal conductivity of pipe is 53 W/m K. Fouling factor on inside and outside are 0.000014 m ² k/W and 0.0015 m ² k/W respectively and convective heat transfer coefficient inside and outside are 2000 W/m ² K and 1000 W/m ² K.	CO5 CO5
(c)	Derive the expression for effectiveness by NTU for counter-flow Heat exchanger.	
Q5 (20 marks)		
(a)	Derive the relation for radiation heat exchange between two infinitely long concentric cylinders.	CO1
(b)	What are the three modes of heat transfer ? Explain the mechanism of energy transfer in three modes of heat transfer.	
(c)	Explain boiling and condensation with types.	